

Release Scope:	v1.2.2 + v1.2.3 (Combined)	Target Branch:	deploy	Architecture:	Multi-Sector & Sub-Categories
Primary Modules:	Complaints, Employees, Dashboard	Database Update:	Sub-Categories, JSON Sectors, Indexes	Report Purpose:	Major Features Handover

PART 1: RELEASE v1.2.2 MAJOR FEATURES

1. High-Speed Searchable AJAX House Dropdown (13,500+ Houses Scalability)
- **Instant Query Scalability:** Replaced static dropdown rendering with GET /admin/houses/search Select2 AJAX search, reducing form load times from seconds to < 50ms across 13,500+ residential properties.
 - **Contextual Auto-Population:** Selecting a house instantly auto-fills Address, Resident Name, Phone, and location coordinates seamlessly in Complaint Registration.
2. Multi-Sector Employee Architecture (JSON Storage & Checkbox Dropdown)
- **Multi-Node Technician Assignment:** Enabled assigning technicians to 1, 2, 3, or more GE Nodes simultaneously using sector_ids JSON array storage while preserving legacy sector relations.
 - **Interactive Checkbox Dropdown:** Custom multi-select UI with **Smart 1-Node Auto-Selection** that automatically pre-checks single available nodes.
3. High-Performance Database Indexing Engine (130,000+ Records)
- **B-Tree Optimization:** Added targeted compound indexes across houses, complaints, employees, spares, and complaint_logs to ensure lightning-fast dashboard analytics and zero page freeze.
4. Multi-Sector Location Permission Governance (LocationFilterTrait)
- **Role-Based Scoping:** Synchronized location permission trait so multi-sector employees are accurately visible to supervisory users across any assigned sector.

PART 2: RELEASE v1.2.3 MAJOR FEATURES

1. Complete Sub-Category Management Ecosystem & Role Permissions
- **Dedicated Sub-Category Module:** Full CRUD interface at /admin/sub-category mapped directly under parent complaint categories.
 - **Independent Permission Governance:** Role-based permissions added in /admin/roles and sidebar navigation for granular administrative security.
2. 3x3 Complaint Registration Grid & Category-Dependent Employee Filtering
- **Modern 3-Row Form Layout:** Streamlined Create/Edit views into an organized 3x3 grid (Category, Sub-Category, Type | Priority, Time, Employee | Description).
 - **Cross-Trade Misassignment Lock:** Assign Employee dropdown dynamically filters technicians to the chosen Category, locked until category selection.
3. Automatic Workflow Transition: Unassigned to 'In Progress'
- **Instant Dispatch State:** Assigning an unassigned complaint to a technician immediately moves status directly to **In Progress** (STATUS_IN_PROGRESS = 0).
4. Universal QR Code Public Feedback Workflow
- **All-Status Feedback Access:** Clients can submit QR feedback on complaints of any status without artificial pre-resolution locks.
 - **Manual Office Resolution:** Feedback submission preserves current complaint status, allowing complaint staff full governance.
5. Dashboard Complaints Modal — 8-Column PDF Export Suite
- **One-Click Landscape PDF:** Generates a clean A4 printable document containing 8 key columns: *CMP-ID, Register Date, House No, Category, Sub Category, Type, Priority, Description*.
6. Frontend & Admin Dashboard Cascading Filter Toolbar
- **Full Cascading Multi-Select Filters:** CMES -> GE -> GE Nodes -> Category -> Sub Category dropdowns with real-time sync, count badges, and empty-state messaging.
7. Emergency vs Normal Priority System & Excel Roster Exports
- **Standardized Priority System:** Color-coded Emergency vs Normal badges across slips, lists, and modals.
 - **1-Click Excel Exports:** Instant roster export buttons on Employee (/admin/employees) and House (/admin/houses) registries.